NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-6106/19E(USAF)
24 October 1990

SUPERSEDING
MIL-R-6106/19D(USAF)
4 January 1989

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC PERMANENT MAGNET DRIVE, 25 AMPERES, SPDT, HERMETICALLY SEALED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this Specification sheet and the issue of the following specification list in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-6106.

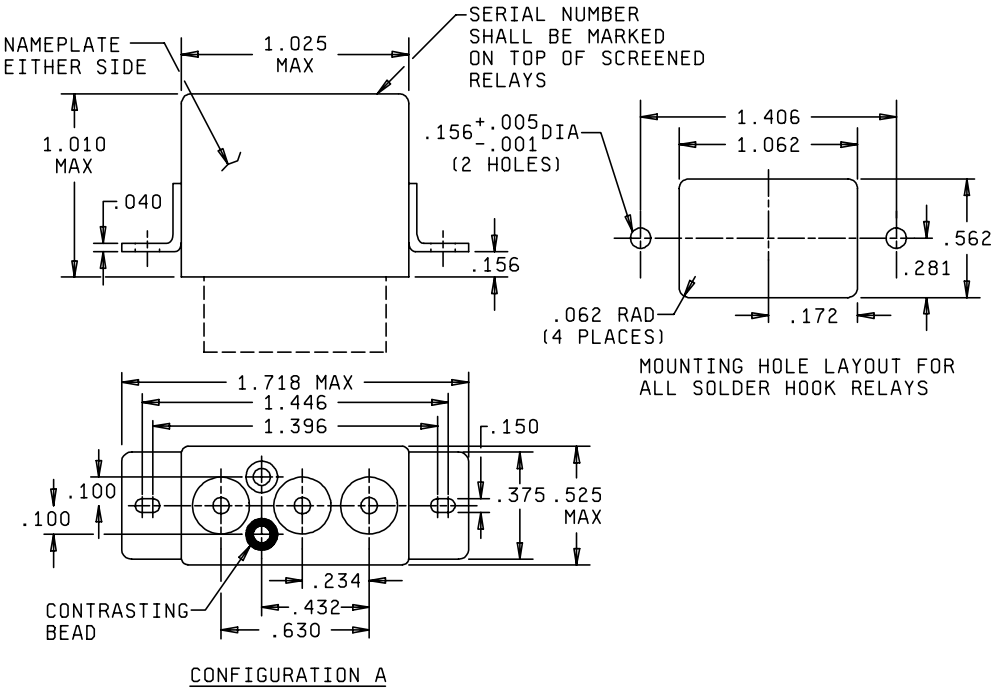
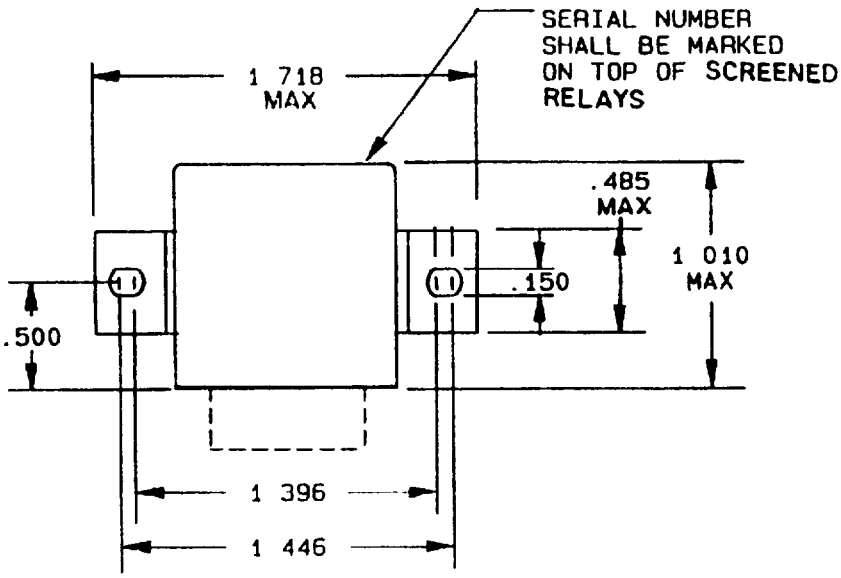
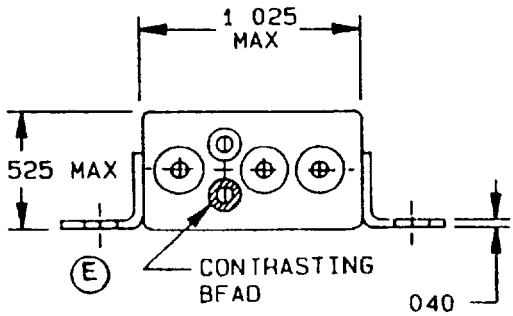


FIGURE 1. Relay, outline drawing.

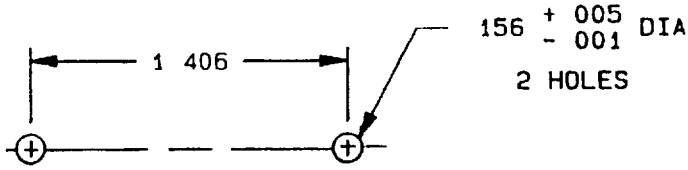
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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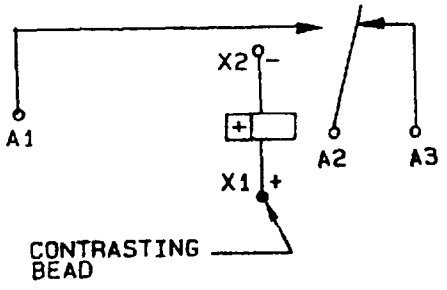
CONFIGURATION B



MOUNTING HOLE LAYOUT

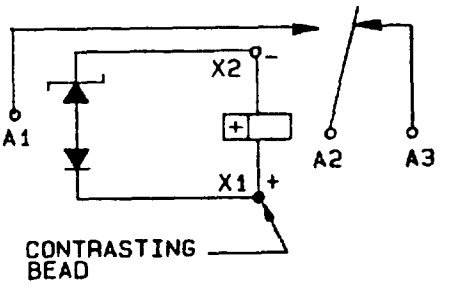
FIGURE 1. Relay, outline drawing - Continued

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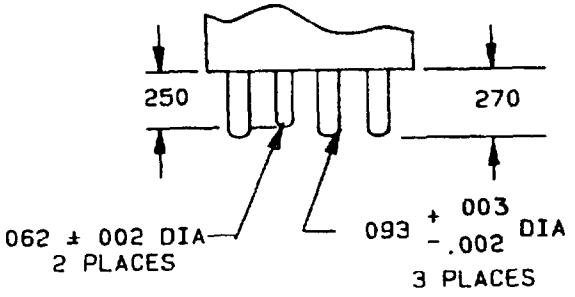
Ⓔ Circuit A

(For details, see table I)



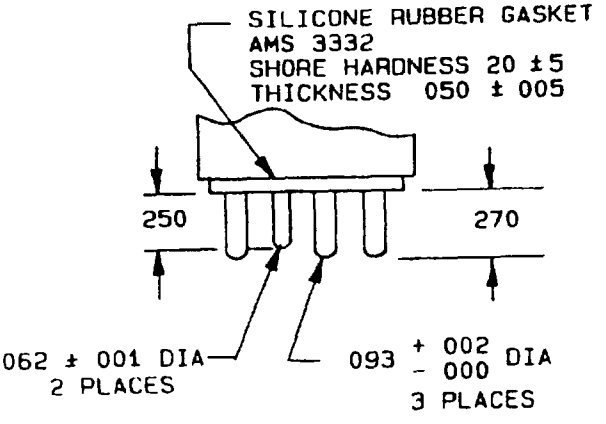
Ⓔ Circuit B

(For details, see table I)



FINISH TINNED

SOLDER PIN TERMINAL

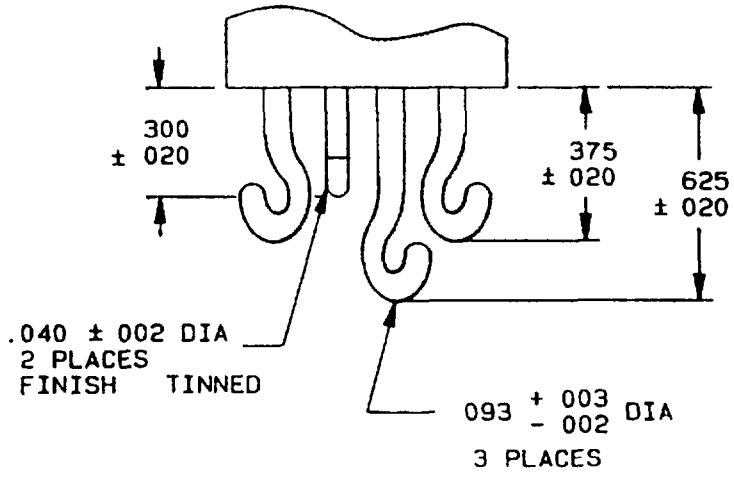


FINISH GOLD PLATE, MIL-G-45204, TYPE II, CLASS 1, UNDERPLATING, NICKEL 50 TO 150 MICROINCHES THICK

SOCKET PIN TERMINAL

FIGURE 1. Relay, outline drawing - Continued

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SOLDER HOOK TERMINAL

Inches	mm	Inches	mm	Inches	mm
001	0 03	156	3.96	.525	13 34
.002	0.05	172	4.37	.562	14.27
.003	0.08	.234	5.94	.625	15.88
.005	0.13	.250	6.35	.630	16.00
.020	0.51	.270	6 86	1.010	25.65
.040	1.02	.281	7.14	1.025	26.04
.050	1 27	.300	7.62	1.062	26.97
062	1 57	375	9.52	1.396	35.46
.093	2 36	432	10 97	1.406	35 71
.100	2 54	485	12.32	1.446	36 73
150	3.61	.500	12 70	1.718	43.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. Relay must not operate or become damaged by reverse polarity. Semiconductors shall not be used for this purpose
6. Hook terminals will accept one no. 12 AWG stranded wire.
7. Permanent magnet drive consists of permanent magnet with its flux path switched and combined with the electromagnetic flux.

FIGURE 1 Relay, outline drawing - Continued.

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(E) TABLE I Mechanical and electrical characteristics.

Dash number	Relay type	Coil 1/2/ suppression	Terminal type	Mounting	Mating socket	Coil circuit	Time current relay characteristic at 25°C
				Configuration			
-003	I	Not applicable	Solder hook	A	Not applicable	A	Not applicable
-004	I	"	Socket, pin	A	MS	A	Not applicable
-005	I 3/	"	Solder, pin	A	Not applicable	A	Applicable
-006	I 3/	"	Solder hook	A	Not applicable	A	"
-007	I 3/	"	Socket, pin	A	MS	A	"
-008	I 3/	"	Solder pin	B	Not applicable	A	"
-009	I 3/	"	Solder hook	B	Not applicable	A	"
-010	I	"	Solder, pin	A	Not applicable	A	"
-011	I	"	Solder hook	A	Not applicable	A	"
-012	I	"	Socket, pin	A	MS	A	"
-013	I	"	Solder, pin	B	Not applicable	A	"
-014	I	"	Solder, hook	B	Not applicable	A	"
-015	I	42 V dc, max	Solder, pin	A	Not applicable	B	"
-016	I	"	Solder, hook	A	Not applicable	B	"
-017	I	"	Socket, pin	A	MS	B	"
-018	I	"	Solder, pin	B	Not applicable	B	"
-019	I	"	Solder hook	B	Not applicable	B	"
-020	I 3/	"	Solder, pin	A	Not applicable	B	"
-021	I 3/	"	Solder hook	A	Not applicable	B	"
-022	I 3/	"	Socket, pin	A	MS	B	"
-023	I 3/	"	Solder, pin	B	Not applicable	B	"
-024	I 3/	"	Solder hook	B	Not applicable	B	"
-025	I	Not applicable	Solder hook	B	Not applicable	A	Not applicable

1/ Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode.

(E)2/ Diodes shall have a peak inverse voltage of 600 V dc minimum when used and shall be JANTX or JANTX screened after 30 December 1980.

(E)3/ Screened relays shall be tested to the group A, type I ER requirements only, additional life testing is not required. For group A inspection, operational reliability shall be performed at the alternate low level run in.

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(E) TABLE II Dash numbers and operating characteristics.

Dash number	Coil data											Time milliseconds (maximum)						
	Coil	Nominal			Volts	Amp	Maximum pickup voltage			Hold 2/ voltage	Dropout voltage 2/	Operate 3/	Release 4/	Contact bounce				Break bounce
														Main		Aux		
		Volts 1/	Freq. (Hz)	Resistance ohms ±10% at 25°C	Max	Max	Normal 2/	High temp.	Continuous current					NO	NC	NO	NC	NO only 5/
-003 through -025	X1 - X2	28	dc	320	29	0.11	18	19.8	22.5	7.0	1.5	10	10	1.0	1.0	---	---	0.10

1/ CAUTION Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.

2/ Over the temperature range.

3/ With nominal coil voltage

4/ From nominal coil voltage.

5/ Applicable to dash numbers -015 through -024 only

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(E) TABLE III. Rated contact load (endurance) (amperes per pole) case grounded.

Type of load	Endurance life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main (Hz)		Main (Hz)		Main (Hz)		Aux (Hz)	
		NO	NC	NO	NC	400	60	400	60	400	60	400	60
Resistive 2/	50	25	25			25							
Inductive	10	12	12			---							
Inductive	20	---	---			15							
Motor	50	10	10			10							
Lamp	50	5	5			5							
Transfer, load 3/	---	---	---			---							
Mechanical life reduced current	200	6	6			6							
Intermediate current 4/													
APPLICABLE ONLY TO DASH NOS. -003, -004, AND -025													
Resistive	25						10						
Inductive	---						---						
Inductive	25						10						
Motor	25						8						
Lamp	---						---						
Transfer, load 3/	---						---						
Mechanical life reduced current	100						2.5						

1/ Absence of value indicates relay is not rated for three phase application

2/ For full rated load maximum temperature and altitude, use no. 12 wire or larger. Solder hook relays to be mounted to limit mounting bracket temperature to 160°C.

3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

(E) 4/ Applicable per MIL-R-6106.

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REQUIREMENTS:

Coil data:

Coil data. See tables I and II.
Duty rating. Continuous.

Operational data.

Contact ratings: See table III.
Operate time: See table II.
Contact drop. 0.150 volt maximum, initial,
0.175 volt maximum, after life test.
Overload current: 1/ 50 amperes dc, 80 amperes ac, 400 Hz.
Rupture current. 2/ 60 amperes dc, 100 amperes ac, 400 Hz.

Physical:

Dimensions and configurations. See figure 1.
Weight: 0.09 pound, 4.1 grams maximum.
(E) Terminal type: See figure 1 and table I.

ENVIRONMENTAL DATA:

Temperature range 3/ -70°C to +125°C
Altitude rating 80,000 feet maximum.
(E) Shock 200 g's.

Duration. 6 ms.
Duration contact opening 10 microseconds maximum.

(E) Vibration - sinusoidal: 30 g's.

Frequency range: 10 to 3,000 Hz.

(E) Random vibration. MIL-STD-202, method 214, test condition IG.

Duration 15 minutes each plane

Acceleration 15 g's

Insulation resistance, initial 100 megohms

After life or environmental tests 50 megohms.

1/ For -003, -004, and -025, 20 amperes ac, 60 Hz.

2/ Not applicable to -003, -004 and -025.

3/ For full rated load maximum temperature and altitude, use no. 12 wire or larger, solder hook relays shall be mounted to limit mounting bracket temperature to 160°C. This note is not applicable to -003, -004, and -025 and the requirements of MIL-R-6106 for temperature and wire size shall apply.

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Dielectric withstanding voltage (sea level):

	Initial	After life test
Coil to case	1,000 V rms	1,000 V rms
All other points	1,250 V rms	1,000 V rms

Dielectric withstanding voltage (altitude): 4/

	80,000 feet	300,000 feet
Coil to case	350 V rms	500 V rms
All other points	350 V rms	500 V rms

Part or Identifying Number (PIN). M6106/19- (dash number from table I).

(E) TABLE IV. Time current relay characteristics at 25°C. 1/ 2/ 3/

Amperes	Duration
30	1 hour
50	5.0 seconds
100	1.2 seconds
250	0.2 second
350	0.1 second

(E) 1/ Applicable to dash numbers -005 through -024.

(E) 2/ CAUTION: Compare with time current characteristics of the associated circuit protective device.

(E) 3/ Dash nos. -005 through -024. Relays shall sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels for the time durations in table IV Separate relays shall be tested at 28 V dc and 115 V ac, 400 Hz. Cooling time between successive applications shall be 30 minutes. The test shall be performed on both normally open and normally closed contacts of each relay. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at conclusion.

4/ Dielectric withstanding voltage rating may be improved by suitable insulation of terminals and wiring after installation. Plug in relays with suitable gasket compressed will have dielectric withstanding voltage capabilities at 80,000 feet over 500 V rms

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Cross reference for Government logistical support. See table V.

Ⓔ TABLE V Cross reference for Government logistical support.

Part or Identifying Number M6106/19-	Support with Part or Identifying Number M6106/19-	Part or Identifying Number M6106/19-	Support with Part or Identifying Number M6106/19-
003	003	015	020
004	004	016	021
005	020	017	022
006	021	018	023
007	022	019	024
008	023	020	020
009	024	021	021
010	020	022	022
011	021	023	023
012	022	024	024
013	023	025	025
014	024		

Ⓔ Supersession data See table VI

Ⓔ TABLE VI Supersession data.

Superseded PIN M6106/19-	New PIN M6106/19-
001	011
002	012

CONCLUDING MATERIAL

Custodian
Air Force - 85

Review activities
Air Force - 99
DLA - ES

Preparing activity
Air Force - 85

Agent.
DLA - ES

(Project 5945-F714)

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